SynDiff Synthetic Benchmark — Live Progress

2026-07-12 19:42:10 CEST

942/942 jobs complete (100.0%)

S = I(X_1;Y|X_2) − I(X_1;Y) in nats; S>0 synergy-dominant, S<0 redundancy-dominant. SynDiff (ours) vs prior estimators.

Dataset construction

Phase 1 — Gaussian (closed-form ground truth)

Scalar base: $X_1, X_2 \sim \mathcal{N}(0,1)$ with $\mathrm{corr}(X_1, X_2) = \rho$; label $Y = X_1 + X_2 + \sigma_y\varepsilon$, $\varepsilon \sim \mathcal{N}(0,1)$. Thus $\mathrm{Var}(Y) = 2 + 2\rho + \sigma_y^2$ and $\mathrm{Cov}(X_i, Y) = 1 + \rho$. To lift to d dimensions the informative scalar is padded with $d-1$ i.i.d. $\mathcal{N}(0,1)$ coordinates and rotated by a fixed per-seed orthogonal matrix Q_i (one per variable). Rotation and independent padding preserve mutual information, so S is unchanged and the closed form carries over. Ground truth is computed from the 3×3 covariance via $I = \frac{1}{2}\log\big(\det\Sigma_A \det\Sigma_B / \det\Sigma_{AB}\big)$ and the analogous conditional form. Cells sweep $\rho \in \{-.9, \ldots, .9\}$, $\sigma_y \in \{.1, 1\}$, $n \in \{1\text{k}, \ldots, 100\text{k}\}$, $d \in \{1, 8, 32, 128\}$.

Phase 2 — discrete U/R/S (mixture-density MC truth)

Latent sources $s_1, s_2, s_3 \sim \mathrm{Unif}\{0, \ldots, Q-1\}$ with $Q = 8$. Encodings $X_i = W_i\,\mathrm{onehot}(s_i) + \sigma_{\mathrm{enc}}\,\mathcal{N}(0, I_{32})$ with orthonormal $W_i \in \mathbb{R}^{32\times 8}$ frozen per seed. Regimes (X_1 always encodes s_1): **UNIQUE** $Y{=}s_1$; **REDUNDANT** $Y{=}s_1$ and X_2 also encodes s_1; **SYNERGY1** $Y{=}(s_1{+}s_2) \bmod Q$ with $X_2 \sim s_2$; **SYNERGY2** $Y{=}(s_3{+}(s_1{+}s_2) \bmod Q) \bmod Q$ with $X_2{=}[\mathrm{enc}(s_2)\,|\,\mathrm{enc}(s_3)] \in \mathbb{R}^{64}$; **INDEPENDENT** $Y{=}s_3$. The λ-dial mixes the SYNERGY1 and REDUNDANT rules per sample (Bernoulli(λ)). Ground truth is the exact $S = I(X_1; X_2, Y) - I(X_1; X_2) - I(X_1; Y)$ obtained by Monte Carlo over the enumerable Gaussian-mixture components (matches the discrete limit as $\sigma_{\mathrm{enc}} \to 0$: $\pm\log Q$ for pure synergy/redundancy, 0 for unique/independent).

Method: SynDiff

Target $S = I(X_1; Y \mid X_2) - I(X_1; Y)$ in nats. By the symmetry of interaction information this equals $I(X_1; X_2 \mid Y) - I(X_1; X_2)$, which the four-mask form below estimates directly. **VP schedule:** $\beta(t) = 0.1 + 19.9t$, $B(t) = \int_0^t \beta = 0.1t + 9.95t^2$, signal $k(t) = e^{-B/2}$, noise $\sigma^2(t) = 1 - e^{-B}$, weight $w(t) = \beta(t)/(2\sigma^2(t))$. A single noise-prediction network $\hat\varepsilon(x_{1,t}, t, \mathrm{cond})$ (*MaskedCondMLP*: Fourier-t features ⊕ per-slot conditioning embeddings each with a learned NULL token; 4×512 SiLU) is trained to denoise X_1 while each conditioning slot (X_2, Y) is independently dropped to its NULL token with probability 0.15. EMA weights (decay 0.999) are used at evaluation. **Four-mask estimator.** Per evaluation triple and per stratified time t (K=16 strata on $[10^{-3}, 1]$) with *one shared* noise draw ε and $x_{1,t} = k(t)x_1 + \sigma(t)\varepsilon$:

$$
\begin{aligned}
e_1 &= \hat\varepsilon(x_{1,t}, t, x_2, y), & e_2 &= \hat\varepsilon(x_{1,t}, t, \varnothing, y),\\
e_3 &= \hat\varepsilon(x_{1,t}, t, x_2, \varnothing), & e_4 &= \hat\varepsilon(x_{1,t}, t, \varnothing, \varnothing),\\
\mathrm{integrand}(t) &= w(t)\big(\|e_1 - e_2\|^2 - \|e_3 - e_4\|^2\big).
\end{aligned}
$$

Per-sample $\hat S = (1 - 10^{-3})\,\overline{\mathrm{integrand}}_t$; the estimate is the mean over 10^4 fresh triples with a 1000-sample bootstrap CI (norms in fp32, no-grad, EMA weights). **Baselines:** *minde_comp* (three separately trained MINDE MI terms composed as above), *MINE* and *InfoNCE* (lower-bound MI, same composition), *batch_pid* (Liang et al. CVX PID, k-means k=20 then max-entropy solve), and *broja*/*plugin* on d=1.

Sweep progress

Phase	Method	Done	Expected
P1	batch_pid	96	96
P1	broja	6	6
P1	infonce	96	96
P1	minde_comp	96	96
P1	mine	96	96
P1	plugin	6	6
P1	syndiff	96	96
P2	batch_pid	90	90
P2	infonce	90	90
P2	minde_comp	90	90
P2	mine	90	90
P2	syndiff	90	90

Accuracy vs ground truth (completed cells, lower error better)

Method	#cells	mean |est−truth|	sign acc
broja	2	0.0680	1.00
plugin	2	0.1301	1.00
minde_comp	62	0.1419	0.81
syndiff	62	**0.2277**	0.89
batch_pid	62	0.3660	0.74
mine	62	0.8132	0.58
infonce	62	3.9327	0.37

SynDiff vs prior — absolute error to truth $|\hat{S} - S|$ (nats)

Each entry is |estimate − truth| in nats; **bold** marks the most accurate method for that cell. '–' means not yet computed.

Cell	truth	**SynDiff**	batch_pid	minde_comp	mine	infonce	plugin
p1_dial_rho0_sy0p1_d8_n20k	+1.963	0.076	1.817	**0.041**	1.332	7.567	–
p1_dial_rho0_sy1_d8_n20k	+0.144	0.008	0.007	**0.001**	0.699	5.459	–
p1_dial_rho0p3_sy0p1_d8_n20k	+1.740	0.063	1.609	**0.062**	1.113	7.200	–
p1_dial_rho0p3_sy1_d8_n20k	+0.007	**0.001**	0.121	0.003	0.610	5.436	–
p1_dial_rho0p5_sy0p1_d8_n20k	+1.477	0.071	1.367	**0.043**	0.952	6.914	–
p1_dial_rho0p5_sy1_d8_n20k	-0.134	**0.001**	0.244	0.005	0.695	5.120	–
p1_dial_rho0p7_sy0p1_d8_n20k	+1.035	0.064	0.955	**0.045**	0.726	6.430	–
p1_dial_rho0p7_sy1_d8_n20k	-0.329	**0.005**	0.417	0.008	0.322	4.861	–
p1_dial_rho0p9_sy0p1_d8_n20k	+0.024	0.057	**0.024**	0.050	0.353	5.345	–
p1_dial_rho0p9_sy1_d8_n20k	-0.610	0.010	0.672	**0.002**	0.196	4.535	–

Cell	truth	**SynDiff**	batch_pid	minde_comp	mine	infonce	plugin
p1_dial_rhon0p3_sy0p1_d8_n20k	+2.047	0.071	1.891	**0.031**	1.323	7.433	–
p1_dial_rhon0p3_sy1_d8_n20k	+0.209	0.010	0.054	**0.001**	0.771	5.572	–
p1_dial_rhon0p5_sy0p1_d8_n20k	+2.023	0.070	1.872	**0.043**	1.228	7.530	–
p1_dial_rhon0p5_sy1_d8_n20k	+0.213	0.016	0.061	**0.001**	0.717	5.555	–
p1_dial_rhon0p7_sy0p1_d8_n20k	+1.896	0.070	1.760	**0.042**	1.207	7.384	–
p1_dial_rhon0p7_sy1_d8_n20k	+0.177	0.019	0.029	**0.005**	0.825	5.552	–
p1_dial_rhon0p9_sy0p1_d8_n20k	+1.473	0.062	1.334	**0.028**	1.008	6.944	–
p1_dial_rhon0p9_sy1_d8_n20k	+0.083	0.029	0.060	**0.006**	0.663	5.283	–
p1_dim_rho0p7_sy1_d128_n20k	-0.329	0.278	0.469	**0.034**	1.524	5.804	–
p1_dim_rho0p7_sy1_d1_n20k	-0.329	**0.000**	0.079	0.006	0.008	0.058	0.137
p1_dim_rho0p7_sy1_d32_n20k	-0.329	0.036	0.463	**0.002**	1.138	5.773	–
p1_dim_rhon0p7_sy1_d128_n20k	+0.177	0.155	**0.039**	0.094	2.115	6.320	–
p1_dim_rhon0p7_sy1_d1_n20k	+0.177	0.009	0.055	0.007	0.006	**0.006**	0.123
p1_dim_rhon0p7_sy1_d32_n20k	+0.177	0.048	0.031	**0.000**	1.262	6.283	–
p1_ns_rho0p7_sy1_d8_n100k	-0.329	**0.003**	0.417	0.007	0.375	4.876	–
p1_ns_rho0p7_sy1_d8_n1k	-0.329	**0.002**	0.417	0.004	0.301	4.888	–
p1_ns_rho0p7_sy1_d8_n20k	-0.329	**0.004**	0.417	0.006	0.323	4.932	–
p1_ns_rho0p7_sy1_d8_n5k	-0.329	**0.003**	0.417	0.008	0.322	4.846	–
p1_ns_rhon0p7_sy1_d8_n100k	+0.177	0.020	0.028	**0.003**	0.825	5.552	–
p1_ns_rhon0p7_sy1_d8_n1k	+0.177	0.017	0.029	**0.006**	0.674	5.515	–
p1_ns_rhon0p7_sy1_d8_n20k	+0.177	0.023	0.029	**0.003**	0.779	5.552	–
p1_ns_rhon0p7_sy1_d8_n5k	+0.177	0.023	0.029	**0.008**	0.630	5.626	–
p2_independent_senc0p1	+0.000	**0.000**	0.142	0.001	0.305	1.923	–
p2_independent_senc0p5	+0.000	**0.000**	0.141	0.000	0.763	2.083	–
p2_independent_senc1	+0.000	**0.000**	0.138	0.000	0.844	2.070	–
p2_lambda0_senc0p1	-2.079	4.088	0.025	0.511	**0.016**	0.112	–
p2_lambda0_senc0p5	-0.730	0.243	0.521	**0.084**	0.279	1.292	–
p2_lambda0_senc1	-0.065	0.065	0.210	**0.017**	0.909	2.019	–
p2_lambda0p25_senc0p1	-0.302	0.136	**0.025**	0.504	0.407	1.979	–
p2_lambda0p25_senc0p5	-0.134	**0.004**	0.160	0.090	0.626	1.928	–
p2_lambda0p25_senc1	-0.016	0.017	0.159	**0.002**	0.764	2.021	–
p2_lambda0p5_senc0p1	+0.814	0.194	**0.019**	0.534	0.676	2.956	–
p2_lambda0p5_senc0p5	+0.272	0.203	**0.040**	0.170	0.826	2.347	–
p2_lambda0p5_senc1	+0.024	**0.020**	0.120	0.020	0.977	2.103	–
p2_lambda0p75_senc0p1	+1.583	0.137	**0.027**	0.479	0.953	3.646	–
p2_lambda0p75_senc0p5	+0.545	0.212	0.169	**0.154**	1.363	2.635	–
p2_lambda0p75_senc1	+0.048	0.044	0.095	**0.037**	0.572	2.106	–
p2_lambda1_senc0p1	+2.079	0.204	**0.013**	0.747	1.249	4.239	–
p2_lambda1_senc0p5	+0.733	0.173	0.256	**0.125**	1.553	2.749	–
p2_lambda1_senc1	+0.067	**0.067**	0.078	0.067	0.862	2.167	–
p2_redundant_senc0p1	-2.079	3.678	0.027	1.084	**0.021**	0.115	–
p2_redundant_senc0p5	-0.730	0.305	0.523	**0.096**	0.194	1.305	–
p2_redundant_senc1	-0.065	0.065	0.199	**0.024**	0.703	2.006	–
p2_synergy1_senc0p1	+2.079	0.172	**0.013**	0.726	1.255	4.113	–
p2_synergy1_senc0p5	+0.733	0.187	0.251	**0.124**	1.651	2.827	–
p2_synergy1_senc1	+0.067	**0.067**	0.078	0.067	1.081	2.121	–
p2_synergy2_senc0p1	+2.079	2.079	**1.362**	2.081	1.457	4.048	–
p2_synergy2_senc0p5	+0.424	0.424	**0.272**	0.424	1.294	2.508	–
p2_synergy2_senc1	+0.010	0.010	0.132	**0.010**	0.758	2.074	–
p2_unique_senc0p1	-0.000	0.000	**0.000**	0.009	0.838	2.042	–
p2_unique_senc0p5	-0.000	**0.000**	0.142	0.005	1.128	2.075	–
p2_unique_senc1	-0.000	**0.000**	0.142	0.002	1.070	2.071	–
# cells most accurate		**17**	11	31	2	1	0